15) $A^2 - B^2 = A^2 - A^2 \dfrac{(1 - V_1/V_2)^2}{(1 + V_1/V_2)^2}$

$$= A^2 \left[\frac{(1 + V_1/V_2)^2 - (1 - V_1/V_2)^2}{(1 + V_1/V_2)^2} \right]$$

$$= \left(\frac{2A}{1 + V_1/V_2} \right)^2 \left(\frac{V_1}{V_2} \right)$$

$$A^2 - B^2 = c^2 (V_1/V_2)$$

$$\boxed{B^2 + (V_1/V_2)\, c^2 = A^2}$$